Code No: 133BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, February -2024
MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Create a mathematical statement involving quantifiers that expresses the statement, "For every real number x, there exists a real number y such that $x + y = 0$". [2]
b) Formulate a compound statement using logical connectives based on a real-world scenario of your choice. Provide the truth table for the compound statement. [3]
c) Define semigroups and monoids. [2]
d) Explain the relationship between Boolean algebra and logic gates. [3]
e) In a group of 10 people, how many ways can a committee of 3 people be selected? [2]
f) How many different arrangements can be made with the letters of the word "BOOKKEEPER"? [3]
g) What is a recurrence relation? [2]
h) Let R be the relation {(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)}, and let S be the relation {(2, 1), (3, 1), (3, 2), (4, 2)}. Find $S \circ R$. [3]
i) What is the sum of the entries in a row of the adjacency matrix for an undirected graph? For a directed graph? [2]
j) How many nonisomorphic simple graphs have five vertices and three edges? [3]

PART – B

(50 Marks)

2.a) i) Translate the following English sentence into predicate logic: "All cats are mammals, but not all mammals are cats". ii) Given the predicates P(x): "x is a prime number" and Q(x, y): "x is less than y", express the statement "There exists a prime number less than every positive integer".
b) i) Define a predicate and a function to represent the statement: "Every person in the room is older than twice their age". ii) Formulate a predicate logic statement to express the following: "There exists a student in the class who has completed all the assignments". [5+5]

OR

3.a) i) Express the statement "There exists a real number such that it is both greater than 1 and less than 2" using predicate logic. ii) Write a statement using nested quantifiers that expresses: "For every positive real number x, there exists a positive real number y such that $x < y$."
b) Prove the statement $\forall x(P(x) \rightarrow Q(x))$ by assuming $\exists x P(x) \wedge \neg Q(x)$ and deriving a contradiction. [5+5]

4.a) Define a function and discuss injective, surjective, and bijective functions.

b) Define a partial order on the set {2,4,6,8}. Determine the least upper bound (join) and greatest lower bound (meet) for elements in the partial order. [5+5]

OR

5.a) Define an algebraic system and distinguish between groups, rings, and fields.

b) i) Show that the set of natural numbers with addition forms a monoid. ii) Prove or disprove: The set of integers under multiplication is a group. [5+5]

6.a) Expand $(x + y + z)^4$ and find the coefficient of x^2y.

b) i) Suppose that a department contains 10 men and 15 women. How many ways are there to form a committee with six members if it must have the same number of men and women? ii) How many bit strings contain exactly eight 0 s and ten is if every 0 must be immediately followed by a 1? [5+5]

OR

7.a) Devise an algorithm for generating all the r-permutations of a finite set when repetition is allowed.

b) How many permutations of the 26 letters of the English alphabet do not contain any of the strings fish, rat or bird? [5+5]

8.a) Find the solution to the recurrence relation $f(n) = 3 f(n/5) + 2n^4$, when n is divisible by 5, for $n = 5^k$, where k is a positive integer and $f(1) = 1$.

b) Find a recurrence relation for the balance B(k) owed at the end of k months on a loan at a rate of r if a payment P is made on the loan each month. [5+5]

OR

9.a) Find a recurrence relation for the balance B(k) owed at the end of k months on a loan of $5000 at a rate of 7% if a payment of $100 is made each month.

b) Solve the recurrence relation $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with $a_0 = -5$, $a_1 = 4$, and $a_2 = 88$. [5+5]

10.a) Let G be a simple graph. Show that the relation R on the set of vertices of G such that uRv if and only if there is an edge associated with {u, v} is a symmetric, irreflexive relation on G.

b) Show that a simple graph G with n vertices is connected if it has more than $(n - 1)(n - 2)/2$ edges. [5+5]

OR

11.a) Show that the property that a graph is bipartite is an isomorphic invariant.

b) Devise an algorithm for constructing Euler circuits in directed graphs. [5+5]

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